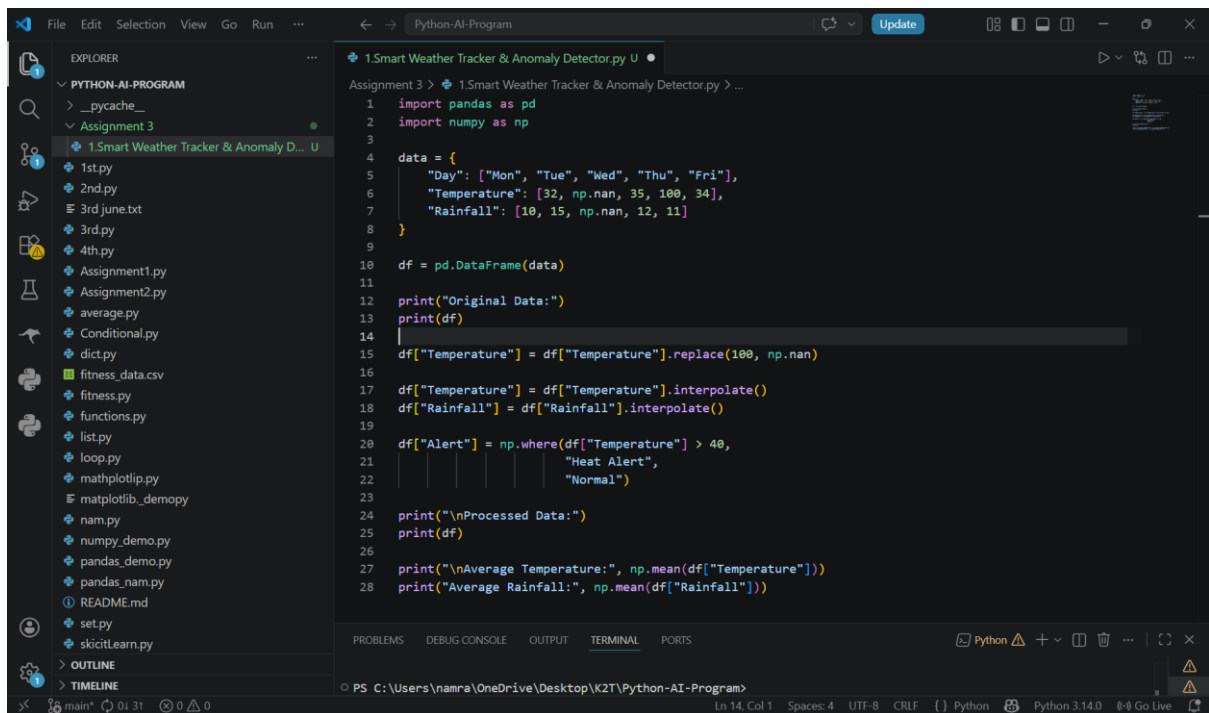
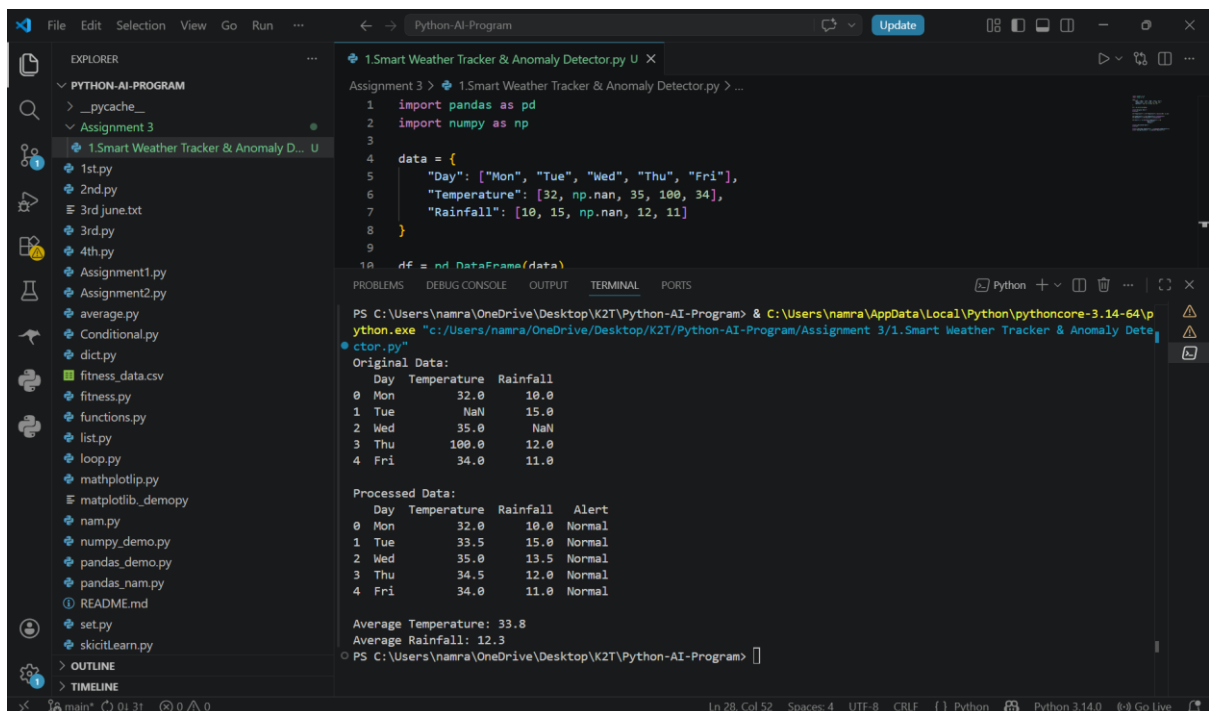


## CODE: 1. Smart Weather Tracker & Anomaly Detector



```
1 import pandas as pd
2 import numpy as np
3
4 data = {
5     "Day": ["Mon", "Tue", "Wed", "Thu", "Fri"],
6     "Temperature": [32, np.nan, 35, 100, 34],
7     "Rainfall": [10, 15, np.nan, 12, 11]
8 }
9
10 df = pd.DataFrame(data)
11
12 print("Original Data:")
13 print(df)
14
15 df["Temperature"] = df["Temperature"].replace(100, np.nan)
16
17 df["Temperature"] = df["Temperature"].interpolate()
18 df["Rainfall"] = df["Rainfall"].interpolate()
19
20 df["Alert"] = np.where(df["Temperature"] > 40,
21                       "Heat Alert",
22                       "Normal")
23
24 print("\nProcessed Data:")
25 print(df)
26
27 print("\nAverage Temperature:", np.mean(df["Temperature"]))
28 print("Average Rainfall:", np.mean(df["Rainfall"]))
```

## OUTPUT:



```
PS C:\Users\namra\OneDrive\Desktop\K2T\Python-AI-Program> & C:\Users\namra\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/namra/OneDrive/Desktop/K2T/Python-AI-Program/Assignment 3/1.Smart Weather Tracker & Anomaly Detector.py"
Original Data:
  Day Temperature Rainfall
0 Mon         32.0      10.0
1 Tue          NaN      15.0
2 Wed         35.0       NaN
3 Thu        100.0      12.0
4 Fri         34.0      11.0

Processed Data:
  Day Temperature Rainfall Alert
0 Mon         32.0      10.0  Normal
1 Tue         33.5      15.0  Normal
2 Wed         35.0      13.5  Normal
3 Thu         34.5      12.0  Normal
4 Fri         34.0      11.0  Normal

Average Temperature: 33.8
Average Rainfall: 12.3
```

